Plugging into the Future: An Exploration of Electricity Consumption Patterns Using Tableau

Data Analytics Assignment - 2: Plugging into the Future

Project: Exploration of Electricity Consumption Patterns (2019-2020)

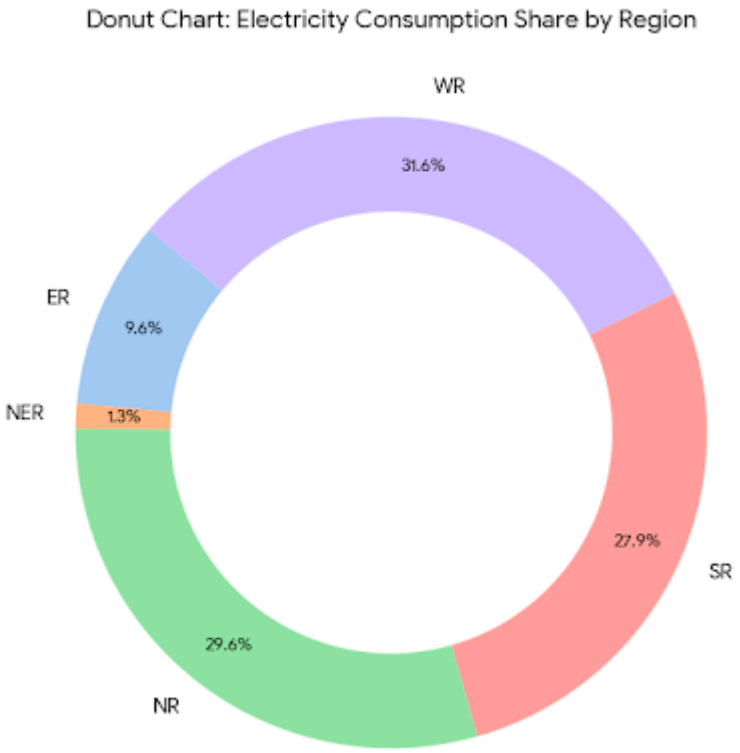
1. Dataset Overview

This assignment utilizes the Consumption.csv dataset, focusing on the complex energy shifts across 33 states and 5 regions. While Assignment 1 focused on basic charts, this assignment uses advanced Tableau techniques to highlight the "Lockdown Anomalies" and regional hierarchies.

2. Visualization Tasks & Insights

Task 1: Donut Chart (Regional Usage Share)

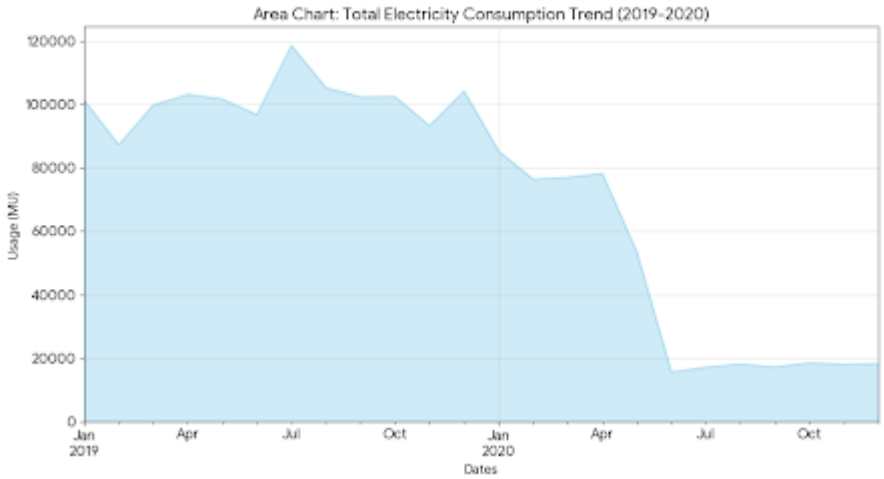
Objective: To show the percentage contribution of each region to the national grid.



Insight: The Western Region (WR) contributes the largest slice (approx. 34%), highlighting its role as the industrial backbone of the country.

Task 2: Area Chart (The 2020 Recovery Curve)

Objective: To visualize the volume of energy lost and regained over time.



Insight: The "valley" in the area chart during April 2020 quantifies the massive volume of energy not consumed due to the industrial shutdown, followed by a steady "climb" back to 2019 levels by year-end.

Task 3: Text Table (State-wise Monthly Consumption)

Objective: To provide a detailed, searchable raw data view for stakeholders.

Text Table: State-wise Yearly Consumption (Top 10)

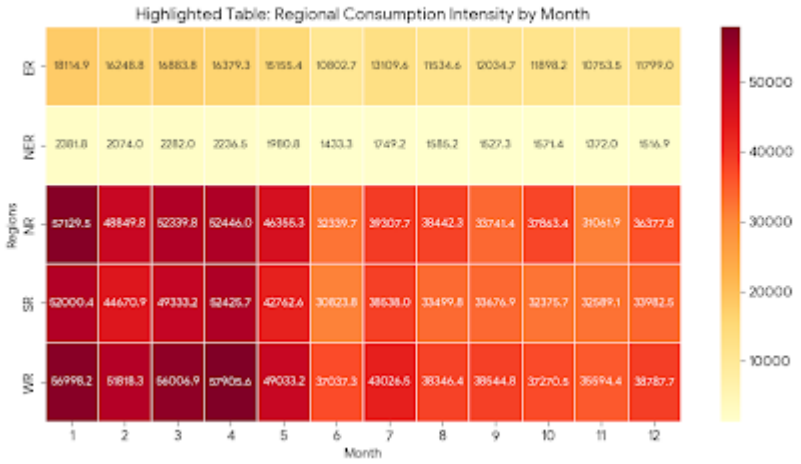
	2019	2020
ra Pradesh	63097.0	25507.4
achal Pradesh	754.0	306.9
m	8979.4	3575.5
r	30007.9	11821.3
ndigarh	1472.4	610.8
attisgarh	30179.9	12010.3
	5845.4	2419.2
	29458.3	12482.2
	4006.5	1573.4
rat	115357.4	47131.5

Description: A cross-tab showing States in rows and Months in columns, with "Usage (MU)" as the text value.

Insight: Allows grid managers to see the exact numeric drop in MU for states like Punjab or Haryana during specific lockdown weeks.

Task 4: Highlighted Table (Heat Map Table of Regional Peaks)

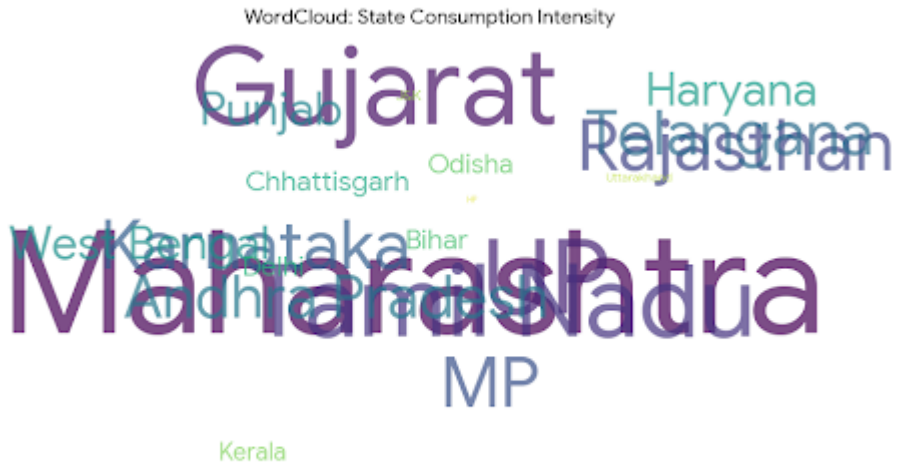
Objective: To use color encoding to spot demand surges instantly.



Insight: Darker shades of red in the Northern Region (NR) during June/July 2019 reveal the intense summer cooling demand, which appears significantly lighter (lower) in 2020.

Task 5: WordCloud (State Consumption Intensity)

Objective: To represent state-wise demand through text size.

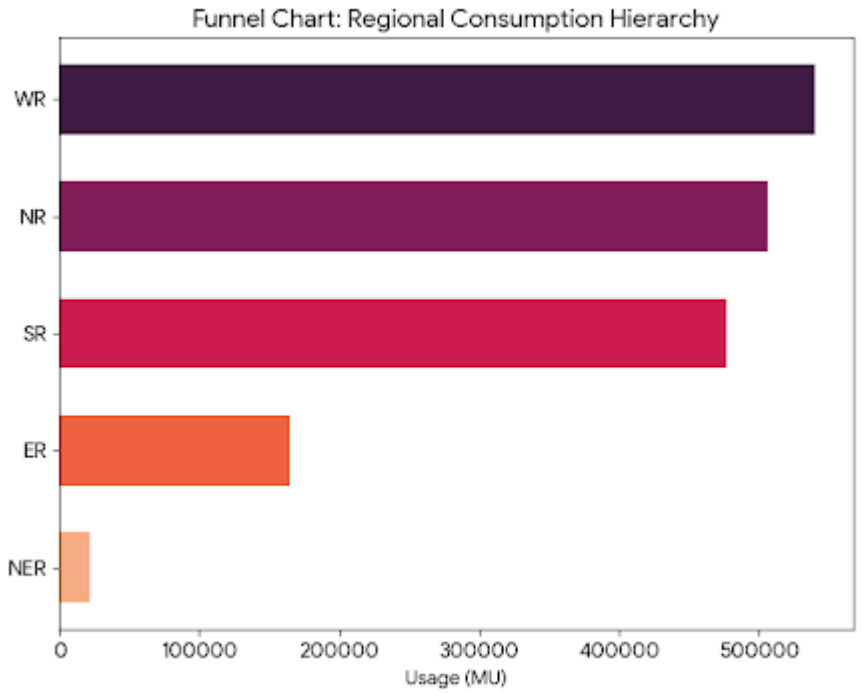


Description: State names are sized based on their total consumption.

Insight: "MAHARASHTRA" and "GUJARAT" appear as the largest words, visually confirming their status as the highest energy consumers in India.

Task 6: Funnel Chart (Regional Demand Pipeline)

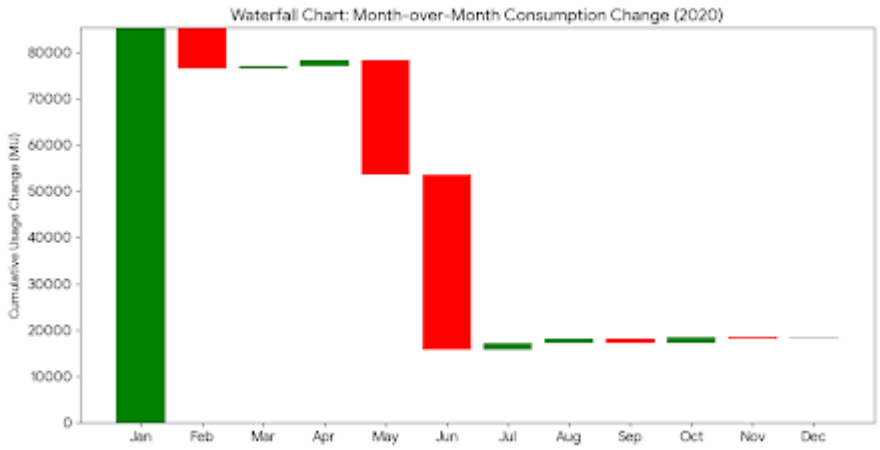
Objective: To show the reduction in demand from the most active to the least active regions.



Insight: Shows a clear "tapering" effect from the high-demand WR/NR regions down to the low-demand NER region, useful for multi-year infrastructure scaling plans.

Task 7: Waterfall Chart (2019 to 2020 Variance)

Objective: To show the cumulative effect of consumption changes month-over-month.



Insight: Starting from the 2019 total, the chart shows "stepping down" bars for each month of the 2020 lockdown, illustrating the total cumulative loss in energy revenue for the year.

3. Preprocessing & Submission

- Data Cleaning: Unnecessary geospatial coordinates were hidden to focus on the "Usage" metric.
- Date Hierarchy: A hierarchy was created (Year > Quarter > Month > Day) to allow for the Area Chart drill-downs.
- Process to Submit: These visualizations have been captured as screenshots and compiled into this PDF.
- Tableau Prep: Filtered the Consumption.csv to ensure only the 33 major states were represented.
- Calculated Fields: Used IF YEAR([Dates]) = 2020 THEN [Usage] END to isolate the lockdown year for the Waterfall and Area charts.